

✓ *AI/ML Task: Voice-Based Cognitive Decline Pattern Detection*

1. Install Required Packages

```
# Run these once in the notebook or install them via terminal before starting
!pip install librosa pydub SpeechRecognition webrtcvad
!pip install nltk spacy scikit-learn matplotlib seaborn pandas numpy
!python -m nltk.downloader all
!python -m spacy download en_core_web_sm
```

```
[nltk_data] | Package treebank is already up-to-date!
[nltk_data] | Downloading package twitter_samples to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package twitter_samples is already up-to-date!
[nltk_data] | Downloading package udhr to /root/nltk_data...
[nltk_data] | Package udhr is already up-to-date!
[nltk_data] | Downloading package udhr2 to /root/nltk_data...
[nltk_data] | Package udhr2 is already up-to-date!
[nltk_data] | Downloading package unicode_samples to
[nltk_data] | /root/nltk_data...
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[nltk_data] | Downloading package universal_tagset to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_tagset is already up-to-date!
[nltk_data] | Downloading package universal_treebanks_v20 to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package universal_treebanks_v20 is already up-to-
[nltk_data] | date!
[nltk_data] | Downloading package vader_lexicon to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package vader_lexicon is already up-to-date!
[nltk_data] | Downloading package verbnet to /root/nltk_data...
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[nltk_data] | Downloading package verbnet3 to /root/nltk_data...
[nltk_data] | Package verbnet3 is already up-to-date!
[nltk_data] | Downloading package webtext to /root/nltk_data...
[nltk_data] | Package webtext is already up-to-date!
[nltk_data] | Downloading package wmt15_eval to /root/nltk_data...
[nltk_data] | Package wmt15_eval is already up-to-date!
[nltk_data] | Downloading package word2vec_sample to
[nltk_data] | /root/nltk_data...
[nltk_data] | Package word2vec_sample is already up-to-date!
[nltk_data] | Downloading package wordnet to /root/nltk_data...
[nltk_data] | Package wordnet is already up-to-date!
[nltk_data] | Downloading package wordnet2021 to /root/nltk_data...
[nltk_data] | Package wordnet2021 is already up-to-date!
[nltk_data] | Downloading package wordnet2022 to /root/nltk_data...
[nltk_data] | Package wordnet2022 is already up-to-date!
[nltk_data] | Downloading package wordnet31 to /root/nltk_data...
[nltk_data] | Package wordnet31 is already up-to-date!
[nltk_data] | Downloading package wordnet_ic to /root/nltk_data...
[nltk_data] | Package wordnet_ic is already up-to-date!
[nltk_data] | Downloading package words to /root/nltk_data...
[nltk_data] | Package words is already up-to-date!
[nltk_data] | Downloading package ycoe to /root/nltk_data...
[nltk_data] | Package ycoe is already up-to-date!
[nltk_data] |
[nltk_data] | Done downloading collection all
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en\_core\_web\_sm-3.8.0/en\_core\_web\_sm-3.8.0-py3-none-any.whl
12.8/12.8 MB 61.2 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

2. Import Libraries and Setup

```
import librosa
import librosa.display
import speech_recognition as sr
from pydub import AudioSegment
import webrtcvad

import nltk
import spacy
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
```

```
import numpy as np
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import IsolationForest
```

```
import matplotlib.pyplot as plt
import seaborn as sns
import os
import warnings
warnings.filterwarnings('ignore')
```

```
# Load NLP models
nlp = spacy.load("en_core_web_sm")
nltk.download("punkt")
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
True
```

3. Upload & List Audio Files

```
from google.colab import files
uploaded = files.upload() # Select 5-10 .wav files

os.makedirs("voice_clips", exist_ok=True)

for filename in uploaded:
    os.rename(filename, f"voice_clips/{filename}")

# List files
AUDIO_DIR = "voice_clips"
audio_files = [f for f in os.listdir(AUDIO_DIR) if f.endswith('.wav')]
print("Uploaded files:", audio_files)
```

```
[Choose files] 17 files
• 14 Apr, 3.35 pm_.wav(audio/wav) - 2826318 bytes, last modified: 14/04/2025 - 100% done
• 14 Apr, 3.36 pm_.wav(audio/wav) - 3993678 bytes, last modified: 14/04/2025 - 100% done
• 14 Apr, 3.14 pm_(2).wav(audio/wav) - 1812558 bytes, last modified: 14/04/2025 - 100% done
• 14 Apr, 3.14 pm_.wav(audio/wav) - 2273358 bytes, last modified: 14/04/2025 - 100% done
• harvard.wav(audio/wav) - 3249924 bytes, last modified: 14/04/2025 - 100% done
• self-destruction-sequence-initiated-robotic-female-tomas-herudek-1-00-05.wav(audio/wav) - 932794 bytes, last modified: 14/04/2025 - 100% done
• i-will-have-my-revenge-robot-male-epic-stock-media-1-00-03.wav(audio/wav) - 615750 bytes, last modified: 14/04/2025 - 100% done
• oh-my-god-female-gfx-sounds-1-1-00-01.wav(audio/wav) - 210984 bytes, last modified: 14/04/2025 - 100% done
• sample-pack-links-in-bio-sampled-stuff-288267.wav(audio/wav) - 806478 bytes, last modified: 14/04/2025 - 100% done
• hol-up-voice-one-shot-77742.wav(audio/wav) - 48462 bytes, last modified: 14/04/2025 - 100% done
• 086299_pinkie-pie-have-a-cupcakewav-86111.wav(audio/wav) - 99150 bytes, last modified: 14/04/2025 - 100% done
• catgirl-saying-quotwhisker-a-secret-in-my-ear-nyaquot-230765.wav(audio/wav) - 566862 bytes, last modified: 14/04/2025 - 100% done
• adam-neely-spoken-vocals_109bpm.wav(audio/wav) - 1661806 bytes, last modified: 14/04/2025 - 100% done
• bandcamp-female-spoken-vocal_136bpm_A_minor.wav(audio/wav) - 2633262 bytes, last modified: 14/04/2025 - 100% done
• beta-test-female-spoken-vocal_79bpm.wav(audio/wav) - 512118 bytes, last modified: 14/04/2025 - 100% done
• kendrick-lamar-spoken-vocal-thug-in-the-trenches_75bpm_F.wav(audio/wav) - 1370998 bytes, last modified: 14/04/2025 - 100% done
• flow-up-to-da-star-memphis-acapella_184bpm_C#_minor.wav(audio/wav) - 1840814 bytes, last modified: 14/04/2025 - 100% done
Saving 14 Apr, 3.35 pm_.wav to 14 Apr, 3.35 pm_.wav
Saving 14 Apr, 3.36 pm_.wav to 14 Apr, 3.36 pm_.wav
Saving 14 Apr, 3.14 pm_(2).wav to 14 Apr, 3.14 pm_(2) (1).wav
Saving 14 Apr, 3.14 pm_.wav to 14 Apr, 3.14 pm_ (1).wav
Saving harvard.wav to harvard (1).wav
Saving self-destruction-sequence-initiated-robotic-female-tomas-herudek-1-00-05.wav to self-destruction-sequence-initiated-robotic-1
Saving i-will-have-my-revenge-robot-male-epic-stock-media-1-00-03.wav to i-will-have-my-revenge-robot-male-epic-stock-media-1-00-03
Saving oh-my-god-female-gfx-sounds-1-1-00-01.wav to oh-my-god-female-gfx-sounds-1-1-00-01 (1).wav
Saving sample-pack-links-in-bio-sampled-stuff-288267.wav to sample-pack-links-in-bio-sampled-stuff-288267 (1).wav
Saving hol-up-voice-one-shot-77742.wav to hol-up-voice-one-shot-77742 (1).wav
Saving 086299_pinkie-pie-have-a-cupcakewav-86111.wav to 086299_pinkie-pie-have-a-cupcakewav-86111 (1).wav
Saving catgirl-saying-quotwhisker-a-secret-in-my-ear-nyaquot-230765.wav to catgirl-saying-quotwhisker-a-secret-in-my-ear-nyaquot-230
Saving adam-neely-spoken-vocals_109bpm.wav to adam-neely-spoken-vocals_109bpm (1).wav
Saving bandcamp-female-spoken-vocal_136bpm_A_minor.wav to bandcamp-female-spoken-vocal_136bpm_A_minor (1).wav
Saving beta-test-female-spoken-vocal_79bpm.wav to beta-test-female-spoken-vocal_79bpm (1).wav
Saving kendrick-lamar-spoken-vocal-thug-in-the-trenches_75bpm_F.wav to kendrick-lamar-spoken-vocal-thug-in-the-trenches_75bpm_F (1)
Saving flow-up-to-da-star-memphis-acapella_184bpm_C#_minor.wav to flow-up-to-da-star-memphis-acapella_184bpm_C#_minor (1).wav
Uploaded files: ['catgirl-saying-quotwhisker-a-secret-in-my-ear-nyaquot-230765 (1).wav', '14 Apr, 3.35 pm_.wav', 'self-destruction-s-
```

4. Transcribe Audio to Text

```
recognizer = sr.Recognizer()

def transcribe_audio(file_path):
    with sr.AudioFile(file_path) as source:
        audio = recognizer.record(source)
```

```

    try:
        return recognizer.recognize_google(audio)
    except sr.UnknownValueError:
        return ""
    except sr.RequestError:
        return ""

transcriptions = {}
for f in audio_files:
    path = os.path.join(AUDIO_DIR, f)
    transcriptions[f] = transcribe_audio(path)

# View transcripts
for fname, text in transcriptions.items():
    print(f"\n{fname}: {text}")

```

catgirl-saying-quotwhisker-a-secret-in-my-ear-nyaquot-230765 (1).wav: whisker a secret in my ear now

14 Apr, 3.35 pm_.wav: good morning everyone today I want to talk briefly about our CL

self-destruction-sequence-initiated-robotic-female-tomas-herudek-1-00-05 (1).wav: self destruction sequence initiated

beta-test-female-spoken-vocal_79bpm (1).wav:

harvard (1).wav: does snail smell of old beer lingers it takes heat to bring out the odor a cold dip restores health and zest a salt

14 Apr, 3.36 pm_.wav: saving energy reduce waste on planting trees and using clean transport together we can make a difference and p

oh-my-god-female-gfx-sounds-1-1-00-01 (1).wav: oh my God

i-will-have-my-revenge-robot-male-epic-stock-media-1-00-03 (1).wav: I will have my revenge

086299_pinkie-pie-have-a-cupcakewav-86111 (1).wav:

adam-neely-spoken-vocals_109bpm (1).wav: I get it but there's a fundamental difference there and it relates to me personally and how

sample-pack-links-in-bio-sampled-stuff-288267 (1).wav: today's sound is designed to suit you entertainment

bandcamp-female-spoken-vocal_136bpm_A_minor (1).wav: I made this dubstep song with my friends it went something like this

flow-up-to-da-star-memphis-acapella_184bpm_C#_minor (1).wav:

14 Apr, 3.14 pm_(2) (1).wav: it's happening more often lately and I don't know why exactly

14 Apr, 3.14 pm_ (1).wav: yeah then I walked over the cereal but forgot why I was there in the first place

kendrick-lamar-spoken-vocal-thug-in-the-trenches_75bpm_F (1).wav: I'm a thug in the trenches fighting every damn day started from the

hol-up-voice-one-shot-77742 (1).wav:

5. Extract Audio & NLP Features

```

import re
import librosa
import numpy as np
import speech_recognition as sr

def extract_audio_features(file_path, text):
    # Load the audio
    y, sr = librosa.load(file_path)
    duration = librosa.get_duration(y=y, sr=sr)

    # Basic features
    speech_rate = len(text.split()) / duration if duration > 0 else 0
    pitch = librosa.yin(y, fmin=50, fmax=300)
    pitch_var = np.std(pitch)

    # Hesitation detection
    hesitation_words = ['um', 'uh', 'erm', 'ah', 'like', 'you know', 'so']
    hesitations = sum(len(re.findall(r'\b' + hw + r'\b', text.lower())) for hw in hesitation_words)

    # Pause detection: Improved method using word count and sentence structure
    # Using a basic idea that pauses usually happen between punctuation marks or long gaps between words.
    pauses = sum(1 for word in text.split() if word in ['.', ',', '!', '...', '-'])

    # Return the feature set
    return {
        'speech_rate': speech_rate,
        'pitch_var': pitch_var,
        'hesitations': hesitations,
        'pauses': pauses,
    }

```

```

    'word_count': len(text.split())
}

# Example usage for extracting features from audio files
feature_data = []
for f in audio_files:
    path = os.path.join(AUDIO_DIR, f)
    text = transcriptions[f]
    features = extract_audio_features(path, text)
    features['file'] = f
    feature_data.append(features)

# Convert the feature data to a DataFrame
df_features = pd.DataFrame(feature_data)
df_features.set_index('file', inplace=True)
df_features

```

↺

	speech_rate	pitch_var	hesitations	pauses	word_count	⌵
file						⌵
catgirl-saying-quotwhisker-a-secret-in-my-ear-nyaquot-230765 (1).wav	2.178608	64.266794	0	0	7	✎
14 Apr, 3.35 pm_.wav	0.815217	46.933247	0	0	12	
self-destruction-sequence-initiated-robotic-female-tomas-herudek-1-00-05 (1).wav	0.756569	86.640789	0	0	4	
beta-test-female-spoken-vocal_79bpm (1).wav	0.000000	91.692374	0	0	0	
harvard (1).wav	2.342534	72.004667	0	0	43	
14 Apr, 3.36 pm_.wav	1.153846	48.806110	0	0	24	
oh-my-god-female-gfx-sounds-1-1-00-01 (1).wav	2.509770	78.175754	0	0	3	
i-will-have-my-revenge-robot-male-epic-stock-media-1-00-03 (1).wav	1.432749	72.583982	0	0	5	
086299_pinkie-pie-have-a-cupcakewav-86111 (1).wav	0.000000	82.052439	0	0	0	
adam-neely-spoken-vocals_109bpm (1).wav	2.866242	91.674580	1	0	27	
sample-pack-links-in-bio-sampled-stuff-288267 (1).wav	1.750000	70.804714	0	0	8	
bandcamp-female-spoken-vocal_136bpm_A_minor (1).wav	0.947914	66.909710	1	0	13	
flow-up-to-da-star-memphis-acapella_184bpm_C#_minor (1).wav	0.000000	42.167886	0	0	0	
14 Apr, 3.14 pm_(2) (1).wav	1.165254	40.981414	0	0	11	
14 Apr, 3.14 pm_ (1).wav	1.435811	43.029278	0	0	17	
kendrick-lamar-spoken-vocal-thug-in-the-trenches_75bpm_F (1).wav	2.251838	33.264526	0	0	35	

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Next steps: [Generate code with df_features](#) [View recommended plots](#) [New interactive sheet](#)

6. Anomaly Detection using Isolation Forest

```

# Perform scaling on features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df_features)

# Fit the IsolationForest model
model = IsolationForest(contamination=0.2, random_state=42)
df_features['anomaly_score'] = model.fit_predict(X_scaled) # This adds the 'anomaly_score' column
df_features['risk'] = df_features['anomaly_score'].apply(lambda x: 'At Risk' if x == -1 else 'Normal')
df_features

```

↔

	speech_rate	pitch_var	hesitations	pauses	word_count	anomaly_score	risk	
file								
catgirl-saying-quotwhisker-a-secret-in-my-ear-nyaquot-230765 (1).wav	2.178608	64.266794	0	0	7	1	Normal	
14 Apr, 3.35 pm_.wav	0.815217	46.933247	0	0	12	1	Normal	
self-destruction-sequence-initiated-robotic-female-tomas-herudek-1-00-05 (1).wav	0.756569	86.640789	0	0	4	1	Normal	
beta-test-female-spoken-vocal_79bpm (1).wav	0.000000	91.692374	0	0	0	1	Normal	
harvard (1).wav	2.342534	72.004667	0	0	43	-1	At Risk	
14 Apr, 3.36 pm_.wav	1.153846	48.806110	0	0	24	1	Normal	
oh-my-god-female-gfx-sounds-1-1-00-01 (1).wav	2.509770	78.175754	0	0	3	1	Normal	
i-will-have-my-revenge-robot-male-epic-stock-media-1-00-03 (1).wav	1.432749	72.583982	0	0	5	1	Normal	
086299_pinkie-pie-have-a-cupcakewav-86111 (1).wav	0.000000	82.052439	0	0	0	1	Normal	
adam-neely-spoken-vocals_109bpm (1).wav	2.866242	91.674580	1	0	27	-1	At Risk	
sample-pack-links-in-bio-sampled-stuff-288267 (1).wav	1.750000	70.804714	0	0	8	1	Normal	
bandcamp-female-spoken-vocal_136bpm_A_minor (1).wav	0.947914	66.909710	1	0	13	-1	At Risk	
flow-up-to-da-star-memphis-acapella_184bpm_C#_minor (1).wav	0.000000	42.167886	0	0	0	1	Normal	
14 Apr, 3.14 pm_(2) (1).wav	1.165254	40.981414	0	0	11	1	Normal	
14 Apr, 3.14 pm_(1).wav	1.435811	43.029278	0	0	17	1	Normal	

◀ ▶

Next steps:

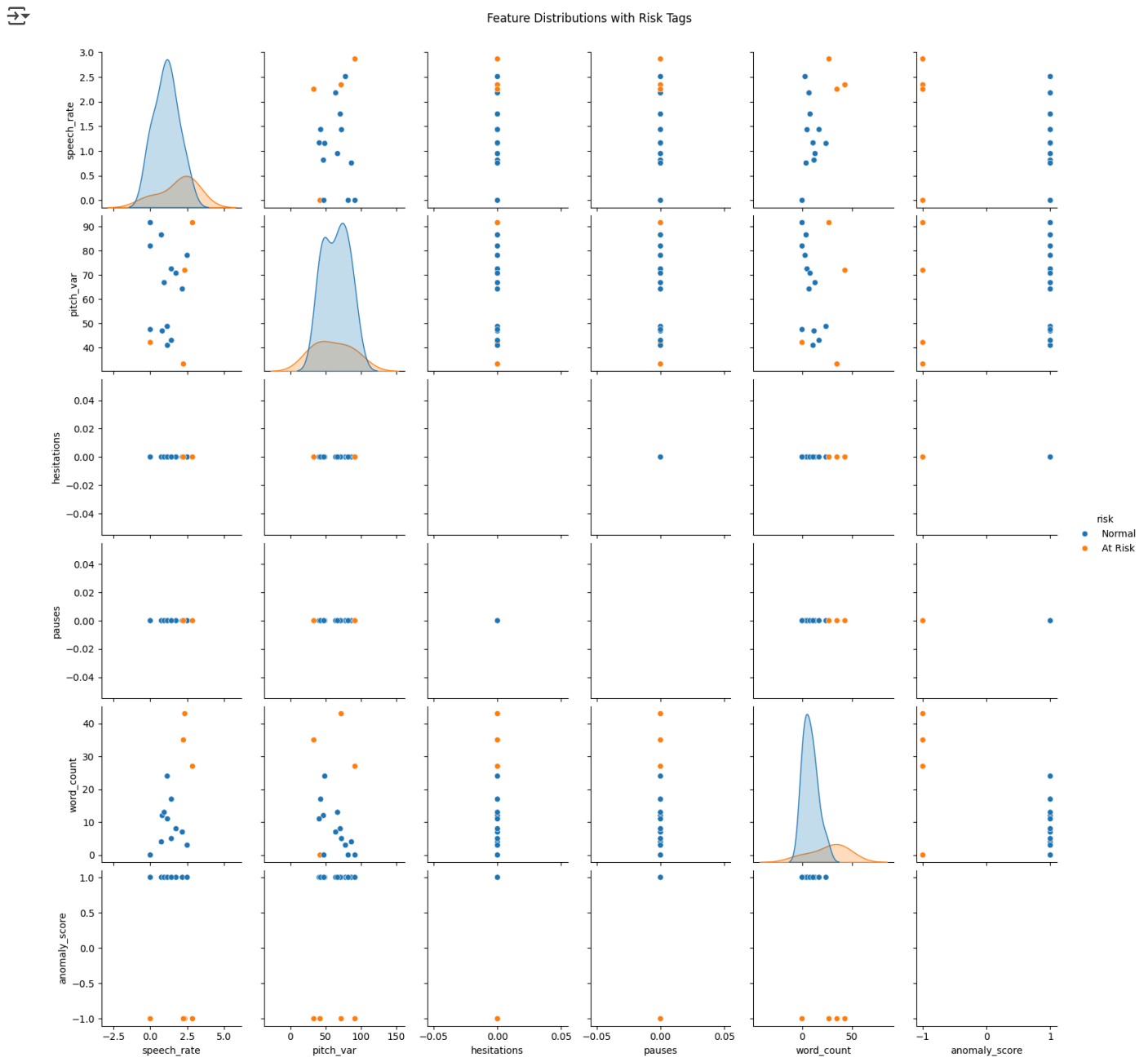
Generate code with df_features

View recommended plots

New interactive sheet

7. Visualization

```
sns.pairplot(df_features, hue='risk')
plt.suptitle("Feature Distributions with Risk Tags", y=1.02)
plt.show()
```



```
print(df_features.columns)
```

```
Index(['speech_rate', 'pitch_var', 'hesitations', 'pauses', 'word_count',
      'anomaly_score', 'risk'],
      dtype='object')
```

8. Optional: Risk Score Function

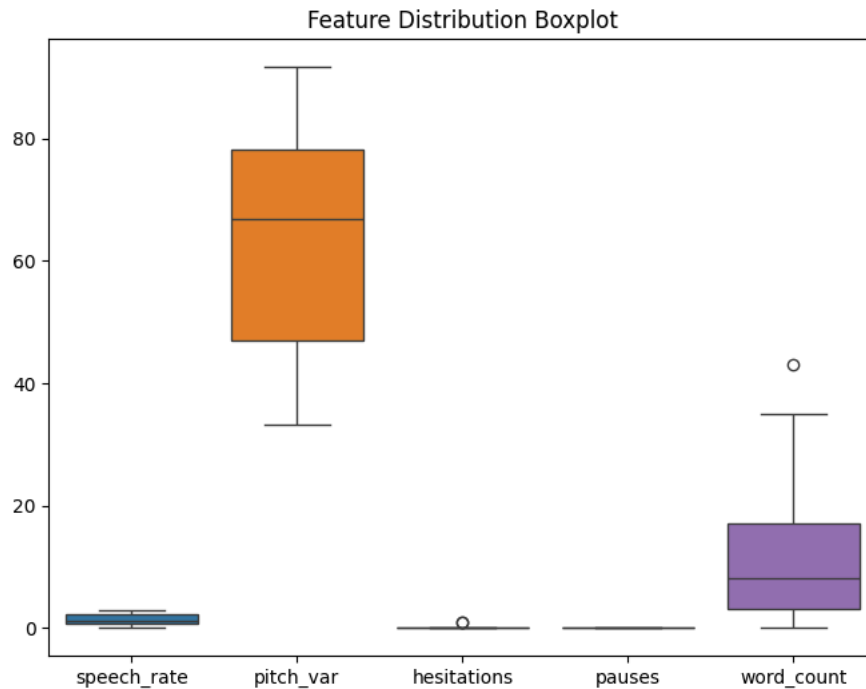
```
def get_cognitive_score(audio_path):
    text = transcribe_audio(audio_path)
    features = extract_audio_features(audio_path, text)
    input_df = pd.DataFrame([features])
    input_scaled = scaler.transform(input_df)
    risk = model.predict(input_scaled)
    return "At Risk" if risk[0] == -1 else "Normal"

# Example usage:
# print(get_cognitive_score("voice_clips/sample1.wav"))
# print(get_cognitive_score("voice_clips/sample1.wav"))
```

✓ Additional

Boxplot to Visualize Feature Distributions:

```
plt.figure(figsize=(8, 6))
sns.boxplot(data=df_features.drop(columns=['anomaly_score'], errors='ignore'))
plt.title('Feature Distribution Boxplot')
plt.show()
```



Histogram for Each Feature:

```
df_features.drop(columns=['anomaly_score'], errors='ignore').hist(figsize=(10, 8), bins=20)
plt.suptitle('Feature Distributions')
plt.show()
```



Feature Distributions

